

HDOC 36

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Batch: 16

1. Algorithm

1. Start the main program.
2. Display the string operations menu with options 1 to 4.
3. Read the user's choice and clear the input buffer to handle string inputs smoothly.
4. Case 1 (Count Characters): Read a string. Iterate through the string character by character until the null terminator \0 is reached, incrementing a counter. Print the counter.
5. Case 2 (Print Each Character): Read a string. Iterate through the string array. For each index before the null terminator, print the character followed by a newline.
6. Case 3 (Count Vowels/Consonants): Read a string. Iterate through the characters. If a character is an alphabetical letter, check if it matches a vowel (both lowercase and uppercase). If it does, increment the vowel counter; otherwise, increment the consonant counter. Print both counts.
7. Case 4 (Exit): Terminate the program.
8. Repeat steps 2-7 until the user selects choice 4.
9. Stop

2. Test-Case Table

TC ID	Scenario	Inputs	Expected Output
TC-01	Count characters	Choice: 1, String: "Hello World"	Number of characters: 11
TC-02	Print characters	Choice: 2, String: "Hi"	H i
TC-03	Count vowels & consonants	Choice: 3, String: "Apple 123"	Vowels: 2 Consonants: 3
TC-04	Exit	Choice: 4	Exiting program...

3. C Program Source Code

```
#include <stdio.h>

void countCharacters(void); void
printCharacters(void);
void countVowelsConsonants(void); void
getString(char str[], int size);

int main() {
    int choice; do {
        printf("\n--- String Operations Menu ---\n"); printf("1. Count characters in a
        string\n");
        printf("2. Print each character of a string on a new line\n"); printf("3. Count vowels and
        consonants in a string\n"); printf("4. Exit\n");
        printf("Enter your choice: "); scanf("%d",
        &choice);

        while (getchar() != '\n');

        switch (choice) { case 1:
            countCharacters(); break;
            case 2:
            printCharacters(); break;
            case 3:
            countVowelsConsonants(); break;
            case 4:
            printf("Exiting program...\n"); break;
            default:
            printf("Error: Invalid choice.\n");
        }
    } while (choice != 4);

    return 0;
}

void getString(char str[], int size) { printf("Enter a string:
"); fgets(str, size, stdin);
int i = 0;
while (str[i] != '\0') {
    if (str[i] == '\n') {
        str[i] = '\0'; break;
    } i++;
}
}
```

```

void countCharacters(void) {
    char str[100];
    getString(str, 100);
    int count = 0;
    while (str[count] != '\0') {
        count++;
    }
    printf("Number of characters: %d\n", count);
}

void printCharacters(void) {
    char str[100];
    getString(str, 100);
    int i = 0;
    while (str[i] != '\0') {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(void) {
    char str[100];
    getString(str, 100);
    int vowels = 0, consonants = 0, i = 0;
    while (str[i] != '\0') {
        char ch = str[i];
        if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
                ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
                vowels++;
            } else {
                consonants++;
            }
        }
        i++;
    }
    printf("Vowels: %d\nConsonants: %d\n", vowels, consonants);
}

```

4. Compilation and Execution

```
(kali@kali)-[~/liak]
$ ./a.out
```

```
— String Operations Menu —
```

1. Count characters in a string
2. Print each character of a string on a new line
3. Count vowels and consonants in a string
4. Exit

Enter your choice: 2

Enter a string: keshav hdoc repo

k
e
s
h
a
v

h
d
o
c

r
e
p
o

```
— String Operations Menu —
```

1. Count characters in a string
2. Print each character of a string on a new line
3. Count vowels and consonants in a string
4. Exit

Enter your choice: 1

Enter a string: keshav hdoc repo

Number of characters: 16

```
— String Operations Menu —
```

1. Count characters in a string
2. Print each character of a string on a new line
3. Count vowels and consonants in a string
4. Exit

Enter your choice: 3

Enter a string: keshav hdoc repo

Vowels: 5

Consonants: 9

```
— String Operations Menu —
```

1. Count characters in a string
2. Print each character of a string on a new line
3. Count vowels and consonants in a string
4. Exit

Enter your choice: 4

Exiting program ...